I am pregnant!
Medical disorders in pregnancy

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Categories

1. Co-incidental or pre-existing
   - Diabetes, renal disease, pre-existing hypertension, autoimmune diseases, cardiac diseases, asthma
2. Arising as a result of pregnancy
   - Hyperemesis gravidarum, gestational hypertension, gestational diabetes, acute fatty liver of pregnancy, acute renal failure or DIC secondary to obstetrics complications
3. Pregnancy increases the risk of some medical disorders or aggravate them
   - Iron deficiency anaemia, thromboembolism

Story of Mary

- Mary was a 26 year old domestic helper who concealed her pregnancy from her employer.
- She had no antenatal care.
- She developed headache, nausea and vomiting and went to the A&E Department
- On physical examination, her BP was 180/100mmHg and she was oedematous. Urine albustix +++. Her abdomen was grossly distended, with fetal movement detected on palpation

Hypertension in pregnancy

- Pre-existing:
  - Hypertension: high BP before 20 weeks
  - Renal disease: proteinuria before 20 weeks (after exclusion of UTI)
- Gestational: (normal BP before 20 weeks, high BP first detected after 20 weeks)
  - Without proteinuria
  - With proteinuria (pre-eclampsia)
- Pre-existing with superimposed pre-eclampsia
  - Pre-existing hypertension, develop proteinuria after 20 weeks

Pre-eclampsia with other organ dysfunction

- Renal insufficiency
- Liver derangement
- Neurological complications
  - eclampsia, altered mental status, blindness, visual scotomata, stroke, clonus, or new-onset severe headaches not responsive to medication and not accounted for by alternative diagnosis
- Haematological complications
  - thrombocytopenia disseminated intravascular coagulation or haemolysis
- Uteroplacental dysfunction
  - fetal growth restriction, abnormal umbilical artery doppler waveform analysis, or stillbirth

Mary’s condition

- In Mary’s case:
- “Unclassified” because she had no antenatal care before. She could have any of the above.
- However, her symptoms were suggestive of pre-eclampsia

Hypertension in pregnancy

- Should drugs be given to lower her blood pressure?
  - Yes
  - Irrespective of the cause of the hypertension, severe hypertension should be controlled to prevent intracerebral haemorrhage

- If so, what drugs can be safely used?
  - Most antihypertensives can be used except ACE inhibitors
  - Pre-eclamptic patients are more sensitive to vasodilators.
    - Use smaller doses than in non-pregnant patients.
  - Rapid lowering of BP may decrease uterine perfusion causing fetal hypoxia

Hypertension in pregnancy

- Any other treatment needed because she is pregnant?
  - Consider prevention of eclampsia
    - Magnesium sulphate
  - Investigations
    - complete blood picture
    - renal/liver function tests
    - coagulation tests
  - Monitor fetal wellbeing when she is stable
  - Consider delivery of the baby

Story of Lin

- Lin was a 25 years old clerk, with insulin dependent diabetes since she was 16.
- She was not very compliant with the treatment and had several episodes of diabetic ketoacidosis in the past.
- She had a steady boyfriend who had been using condom for contraception until about 6 months ago.
- They were getting married soon and thought that it did not matter if she got pregnant.
- She was admitted one day because of hypoglycaemia.

Story of Lin

- Her LMP was 10 weeks ago and pregnancy test was positive.
- She was very happy that she was pregnant.
- However, her blood glucose became more difficult to control after she was pregnant and… there could be other problems.

Diabetes mellitus

Effect of pregnancy on pre-existing DM

- Blood glucose more difficult to control
  - fasting hypoglycaemia
  - antagonising effect of placental hormones on insulin
- Increased stress on the cardiovascular and renal system
- Progression of diabetic retinopathy

Diabetes Mellitus

Effects on Pregnancy

- Maternal aspects:
  - Increased risk of the following complications
    - Pre-eclampsia
    - Urinary tract infection
    - Preterm labour
  - Increased incidence of caesarean section and instrumental delivery

Diabetes Mellitus

- Fetal/Neonatal aspects:
  - Increased risk of congenital malformations
    - Neural tube
    - skeletal
    - Cardiac
    - Renal
    - Gastrointestinal

- Preterm delivery
- Large-for-gestational age
- Asphyxia and birth trauma

- Metabolic complications
- Sepsis
- Jaundice and phototherapy
- Respiratory complications

- Long term consequences on the offspring
  - Fetal programming effect on metabolic and cardiovascular diseases

Diabetes Mellitus

Management Issues

Starts before pregnancy:

- good glycaemic control before pregnancy
- hyperglycaemia is teratogenic
- incidence of major congenital abnormality directly proportional to HbA1c level

Diabetes Mellitus

Management Issues

During pregnancy:

- strict glycaemic control
- oral hypoglycaemics are not contraindicated
  - Strict glycaemic control more difficult to achieve
- insulin usually needed
- monitor for maternal and fetal complications

Story of Lin continued

- Endocrinologist
  - readjusted Lin’s insulin dosage
- Dietitian
  - reviewed with Lin her diet
  - advised extra caloric intake to cover the baby’s need
- DM nurse
  - instructed Lin on how to adjust the insulin dosage on a day-to-day basis
  - gave her a hot-line number for contact if she encountered any difficulties in the glycaemic control
- Detailed fetal ultrasound examination
  - to exclude congenital abnormalities

Story of Lin continued

- Regular follow up by obstetrician and endocrinologist
- Regular ultrasound examination to check on fetal growth
- Labour was induced at 39 weeks
  - tight blood glucose control using insulin-dextrose drip and potassium replacement
- She delivered a 2.9kg baby girl
  - early feeding
  - regular blood glucose monitoring by dextrostix
- Except a few episodes of mild hypoglycaemia, and NNJ requiring phototherapy, the baby remained well and was discharged with Lin on day 5.

Gestational diabetes

- A woman who is not diabetic may become diabetic during pregnancy or diabetes may be first detected during pregnancy
  - gestational diabetes

- She and her baby faces similar problems but usually to a milder degree

- Management will be similar
  - Good glycaemic control is the cornerstone of management
    - by diet +/- insulin

The story of Tina

- Tina was 35 years old
- She had Grave's disease, with relapse after antithyroid treatment
- She opted to have radioactive iodine treatment when she was 30 years old and needed thyroxine replacement 100mcg daily.
- She was not very compliant with the thyroxine replacement but was clinically euthyroid all along
- She got pregnant and at 12 weeks pregnancy, the TSH checked was 6.8 mu/L, normal FT4. She was clinically euthyroid
- Do we need to increase the thyroxine replacement?

Thyroid disease and pregnancy

- Transient biochemical hyperthyroidism
  - no clinical features of thyrotoxicosis
  - can occur in early pregnancy
    - especially in women with hyperemesis gravidarum

- Thyroid disease, especially autoimmune thyroiditis occur more frequently in postpartum period

- Thyroxine has no known adverse effect on the pregnancy unless clinically hyperthyroid

- Subclinical hypothyroidism (high TSH, normal FT4)
  - Associated with preterm labour, increase incidence of delayed neurological development in offsprings
    - observational data only, effectiveness of routine screening and treatment not known yet

- For hyperthyroidism
  - RAI contraindicated in pregnancy
  - Propylthiouracil and carbimazole are commonly used
    - No known teratogenicity
    - PTU – cross less through placenta; risk of liver failure
    - Both can theoretically caused neonatal hypothyroidism, usually reversible

- In Grave's disease
  - autoantibodies can cause neonatal thyrotoxicosis (but rare)

The story of Tina

- High TSH:
  - Poor compliance to thyroxine replacement
  - True subclinical hypothyroidism
    - check compliance, reinforce importance of compliance
    - increase dosage if patient compliant but TSH still elevated

The Story of Tung

- Tung was a 30 years old housewife.
- She had SLE presenting with lupus nephritis when she was 24.
- She had an unplanned pregnancy at age 27 when she had a flare up of SLE.
  - Her renal function rapidly deteriorated at 12 weeks, with heavy proteinuria
  - Pregnancy was terminated.
- Her SLE was subsequently well controlled and she was maintained on prednisolone 5mg daily.
- Her renal function was normal but she still had mild proteinuria of around 600mg per day.
- She wanted to have a baby because her SLE was under control with treatment.

Effect of pregnancy on autoimmune disorders

- Pregnancy is associated with changes in the immune system.
  - Some autoimmune diseases may worsen, some may improve during pregnancy.

- Pregnancy may further stress the end-organ/system affected by the autoimmune disease e.g. kidney

- In general, pregnancy is contraindicated during an acute flare of the disease and is safest during remission

Effect of autoimmune diseases on pregnancy

- Autoantibodies can cross the placenta and affect the baby or affect placentation:
  - Fetal thrombocytopenia in maternal ITP
  - Congenital heart block induced by maternal anti-Ro antibodies
  - Neonatal lupus
  - Lupus anticoagulant causing IUGR and pregnancy loss

Effect of autoimmune diseases on pregnancy

- End organ/system disease can adversely affect the pregnancy:
  - Autoimmune thyroiditis causing hypothyroidism can affect fetal neurological development
  - Renal problem and hypertension can cause IUGR, increase prevalence of superimposed pre-eclampsia
  - ITP can cause excessive bleeding during delivery

Effects of treatment of autoimmune diseases on pregnancy

- Steroids are often used to treat autoimmune diseases
  - IUGR, fetal cleft lip (small risk)
- Immunosuppressants are potentially teratogenic
  - Aziathioprine relatively safe
- Cytotoxic drugs used may be teratogenic
  - Methotrexate is contraindicated in pregnancy

Should Tung try to have a baby?

- Yes, her disease is well controlled on low dose steroids
- Need to check antibodies which may specifically affect the pregnancy, e.g. anti-Ro, lupus anticoagulant, anticardiolipin
- Prepared for a small to moderate increase risk of pre-eclampsia, IUGR and pregnancy loss
- Need joint management by rheumatologist and obstetricians during pregnancy

The story of Lian

- Lian was a 17 years old student, with Tetralogy of Fallot diagnosed at birth.
- She had corrective surgery done in childhood and her right ventricular function was good on digoxin. However, she had pulmonary regurgitation and was scheduled to undergo pulmonary valve replacement after her examinations. She was also on frusemide.
- She had an unplanned and unwanted pregnancy and requested termination of pregnancy. However, the uterus was 26 weeks size already and ultrasound examination confirmed that the fetus was 26 week size.

Cardiac conditions in pregnancy:

- ***Effects of Pregnancy on Cardiac Conditions:***
- Heart failure can occur during pregnancy in a woman with well-compensated cardiac disease before pregnancy:
  - Physiological increase in cardiac output in pregnancy
  - Antenatal anaemia
  - Drug use in pregnancy
    - beta-adrenergics for arresting preterm labour
  - Stress and physical demand during labour and delivery
  - Excessive intravenous fluids

Cardiac conditions in pregnancy:

- *Effects of Pregnancy on Cardiac Conditions:*

- In the postpartum period, decrease in the uterine blood volume may increase the effective circulatory volume leading to volume overload.
  - Precipitate heart failure
    - often right heart failure
  - Increase R to L shunting especially in the presence of pulmonary hypertension

Cardiac conditions in pregnancy:

- ***Effects of Cardiac Conditions on Pregnancy :***

- Fetal size is smaller and preterm labour is more frequent because:
  - Normal physiological increase in the cardiac output may be limited by the cardiac condition
  - Hypoxia in cyanotic heart disease

Cardiac conditions in pregnancy:

- *Effects of Treatment of Cardiac Conditions on Pregnancy :*
- Diuretics may limit the normal physiological volume expansion of pregnancy, leading to intrauterine growth restriction
- Warfarin increases the risk of congenital abnormalities (warfarin embryopathy), fetal loss
- ACE inhibitors increase fetal loss, cause oligohydramnios and decrease fetal renal perfusion

Cardiac conditions in pregnancy:

- *Effects of Pregnancy on Treatment of Cardiac Conditions :*

- Teratogenic drugs
  - may need to be substituted before pregnancy and in the first trimester

- Pharmacokinetic changes in pregnancy affect the serum level of drugs
  - digoxin, warfarin

- Careful readjustment of dosage may be needed during pregnancy

- Other forms of treatment: cardiac catheterization (X-ray exposure), open heart surgery (hypotension, hypothermia, hypoxia) often carry high risks to mother and baby during pregnancy

Cardiac conditions in pregnancy:

- Other considerations :
  - Prevention of bacterial endocarditis during delivery: antibiotics needed for mechanical heart valves or grossly damaged valves but not for most other cases
  - Prevention of thromboembolism
  - Time of delivery and mode of delivery: availability of multidisciplinary specialists support
    - In general, spontaneous onset of labour, vaginal delivery, +/- epidural analgesia carries lowest risks but needs to be individualized

Lian's problems

- Social

- Effects of pregnancy on her condition
  - Will she develop heart failure?

- Effects of her condition on her pregnancy
  - At risk of preterm labour, IUGR?

- Effects of the treatment on her pregnancy
  - Continue digoxin and frusemide? Need to change dosage?

- Effects of pregnancy on her treatment
  - Need to postpone pulmonary valve replacement

Lian's problems

- Other considerations:
  - Mode of delivery, timing of delivery
  - Future contraceptive advice

Lian's story continued

- Lian decided to keep the baby and postponed taking her exams till next year.
- Her mother was supportive and agreed to look after her and her baby.
- A joint case conference with obstetricians, paediatric and adult cardiologists, anaesthesiologists was held to plan the management
- She was continued on digoxin and frusemide.
- Paediatric cardiologists assessed her every 4 weeks till 32 weeks and then every 2 weeks.
- Frusemide was increased at 32 weeks gestation because of progressive shortness of breath.

Lian's story continued

- The fetal growth was monitored every 4 weeks with ultrasound examination and the fetal growth was along the 10th centile
- Labour was induced at 38 weeks, with epidural analgesia for pain relief
- No antibiotics cover needed
- Normal vaginal delivery after 5 hours of labour, baby boy 2.7 kg, healthy
- Barrier contraception advised and pulmonary valve replacement to be scheduled

The story of Hong

- Hong was 28 years old, with grand mal epilepsy, well controlled on phenytoin since she was 16 years old
- She was married for 2 years and decided to get pregnant. She stopped phenytoin once she knew that she was pregnant because she was worried about fetal abnormalities associated with use of phenytoin
- She had 3 attacks in one month after stopping phenytoin

Epilepsy and pregnancy

- Many antiepileptic drugs are teratogenic
  - cleft lip and palate, Neural tube defect
- Should antiepileptic drugs be stopped?
- If epilepsy cannot be controlled without drugs, do not stop the drugs.
  - Consider monotherapy if possible.
- Folic acid supplement decrease the incidence of congenital abnormalities associated with antiepileptic drugs
- Detailed prenatal ultrasound examination to assess fetal anatomy

Jane’s story

- Jane was a 39 years old smoker. She was 80kg.
- She had threatened miscarriage at 12 weeks pregnancy was advised to have bed rest.
- After one week in bed, she developed mild shortness of breath and went to see her doctor who prescribed some cough remedy for her.
- She suddenly collapsed at home on the same day.
- What happened?
  - Postmortem examination showed massive pulmonary embolism

Venous thromboembolism and pregnancy

- Risk of thromboembolism is increased in pregnancy:
- Virchow's triad (risk factors for thromboembolism) are all present during pregnancy:
  - Immobility
    - relatively less mobile during pregnancy
  - Hypercoagulability
    - increase in pregnancy
    - further increase in the postpartum period
    - viscosity may be increased in hyperemesis
  - Obstruction to blood flow
    - gravid uterus

Venous thromboembolism and pregnancy

- Assessment of risk:
- Pre-existing
  - Demographic
    - advance age, obesity, ethnicity
  - Smoking
  - Congenital or acquired thrombophilia
  - History of thromboembolism
    - family history

- Current
  - Pregnancy and treatment related
    - pre-eclampsia, hyperemesis, bed rest, Caesarean delivery, postpartum genital tract infection
  - Concurrent to pregnancy
    - infection, immobilization due to injury, long haul flights

Venous thromboembolism and pregnancy

- Prevention of thromboembolism in pregnancy:
  - Minimize immobility
    - do not prescribe bed rest for conditions where this is not proven to be effective
    - do not prescribe bed rest for threatened miscarriage, IUGR
  - Adequate hydration
  - Special stockings

- Women at increased risk of developing thromboembolism during pregnancy and postpartum should receive prophylactic anticoagulants
  - LWMH or heparin
  - Warfarin
    - Crosses the placenta and has long half life
      - Second trimester, early third trimester and postpartum period

Take home messages

- Obstetrics conditions can mimic medical conditions
  - e.g. pre-eclampsia mimicking hypertensive encephalopathy
- Women with chronic medical diseases can get pregnant
  - pre-pregnancy counselling, contraceptive advice, careful use of drugs in women in reproductive age group
- Effects of pregnancy on pre-existing medical disease and its treatment
- Effects of medical disease and its treatment on the pregnancy
- Consider both mother and the baby
- Prevalence of some medical diseases increase during pregnancy of the postpartum period
  - E.g. venous thromboembolism

The End

Treat the patient, not the disease.